

Abstract Linear Algebra

Prof. Lanier

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Pick up a handout. Try those warm-up problems.

Next up:

- This week: finish Ch 5; Ch 6.1-2 and Kalman article for SVD
- Quiz 10 today (orthogonality, Gram–Schmidt)
- Problem session / office hours today, 3-5pm, Ryerson 177
- HW7 debrief/pre-brief on Sunday, 3-4 in Cobb 115
- HW7 due next Monday
- Make a move on flex! And come by office hours! I'll send a “flex accounting” on Saturday.
- Final Exam review sessions: TBA (packet)
TBA (2021 final)
- Final Exam: Both exams will be held in our classroom, Ryerson 276.
Section 45 (11:30am section) - Thursday, May 25, 5:30-7:30pm
Section 55 (12:30pm section) - Wednesday, May 24, 7:30-9:30am

Ch. 5 & 6: “Now with orthogonality!”

Row reduction:

apply row operations to make
an upper triangular matrix

LU decomposition:

$$A=LU$$

diagonalization:

$$A = PDP^{-1}$$

when A has n lin ind eigenvectors

matrix equivalence:

$$A= Q“I” P, \text{ with } Q \text{ \& } P \text{ invertible}$$

Gram–Schmidt:

apply column operations to make
an orthogonal matrix

QR decomposition:

$$A=QR$$

Spectral Theorem:

$$A = PDP^{-1}$$

with P unitary (orthogonal, for $\mathbb{R}$)

when A is self-adjoint (symmetric for $\mathbb{R}$)

$$A = A^*$$

Singular Value Decomposition

$$A= Q“D”P, \text{ with } Q \text{ \& } P \text{ orthogonal}$$

Upper triangular (Schur) representation (Thms. 6.1.1 & 6.1.2): Let A be the matrix of an operator on an inner product space. Then A can be represented as $A = UTU^*$, where U is a unitary matrix and T is an upper triangular matrix. In the case of real square matrices, any real square matrix A with all real eigenvalues can be represented as $A = UTU^{-1}$, where U is a real orthogonal matrix and T is a real upper triangular matrix.

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Schur representation: Every linear transformation in an inner product space is a stretch-and-shear in some orthonormal basis.

Every square matrix is similar to an upper triangular matrix.

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Every square matrix is similar to an upper triangular matrix.

Pf: induct on dimension

For dim 1, every matrix is upper triangular.

For dim n , rotate to change basis so that one of the new basis vectors is an eigenvector.

The induction hypothesis takes care of the rest.

$$\begin{pmatrix} * & * & * & * \\ * & * & * & * \\ * & * & * & * \\ * & * & * & * \end{pmatrix}$$

The steps
of the proof
of Schur

$$\begin{pmatrix} * & * & * & * \\ * & * & * & * \\ * & * & * & * \\ * & * & * & * \end{pmatrix} = \begin{pmatrix} & & & \\ & & & \\ & & & \\ & & & \end{pmatrix} \begin{pmatrix} & & & \\ & & & \\ & & & \\ & & & \end{pmatrix}^{-1}$$

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$$\begin{pmatrix} * & * & * & * \\ * & * & * & * \\ * & * & * & * \\ * & * & * & * \end{pmatrix} = \begin{pmatrix} | & & & \\ v_1 & & & \\ | & & & \\ | & & & \end{pmatrix} \begin{pmatrix} \lambda_1 & & & \\ 0 & & & \\ 0 & & & \\ 0 & & & \end{pmatrix} \begin{pmatrix} | & & & \\ v_1 & & & \\ | & & & \\ | & & & \end{pmatrix}^{-1}$$

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The steps
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$$\begin{pmatrix} \$ & \$ & \$ \\ \$ & \$ & \$ \\ \$ & \$ & \$ \end{pmatrix} = \begin{pmatrix} | & \\ v_2 & Q_2 \\ | & \end{pmatrix} \begin{pmatrix} \lambda_2 & @ & @ \\ 0 & @ & @ \\ 0 & @ & @ \end{pmatrix} \begin{pmatrix} | & \\ v_2 & Q_2 \\ | & \end{pmatrix}^{-1}$$

$$\begin{pmatrix} * & * & * & * \\ * & * & * & * \\ * & * & * & * \\ * & * & * & * \end{pmatrix} = \begin{pmatrix} | & \\ v_1 & \\ | & \\ | & \end{pmatrix} Q_1 \begin{pmatrix} \lambda_1 & \$ & \$ & \$ \\ 0 & \$ & \$ & \$ \\ 0 & \$ & \$ & \$ \\ 0 & \$ & \$ & \$ \end{pmatrix} \begin{pmatrix} | & \\ v_1 & \\ | & \\ | & \end{pmatrix} Q_1^{-1}$$

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$$\begin{pmatrix} \lambda_1 & \$ & \$ & \$ \\ 0 & \$ & \$ & \$ \\ 0 & \$ & \$ & \$ \\ 0 & \$ & \$ & \$ \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & | & \\ 0 & v_2 & Q_2 \\ 0 & | & \end{pmatrix} \begin{pmatrix} \lambda_1 & @ & @ & @ \\ 0 & \lambda_2 & @ & @ \\ 0 & 0 & @ & @ \\ 0 & 0 & @ & @ \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & | & \\ 0 & v_2 & Q_2 \\ 0 & | & \end{pmatrix}^{-1}$$

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$$= \begin{matrix} \dots \\ Q \begin{pmatrix} \lambda_1 & \# & \# & \# \\ 0 & \lambda_2 & \# & \# \\ 0 & 0 & \lambda_3 & \# \\ 0 & 0 & 0 & \lambda_4 \end{pmatrix} Q^{-1} \end{matrix}$$

Ch. 5 & 6: “Now with orthogonality!”

Row reduction:

apply row operations to make
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LU decomposition:

$$A=LU$$

diagonalization:

$$A = PDP^{-1}$$

when A has n lin ind eigenvectors

matrix equivalence:

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By Schur, $A = UTU^{-1}$ with T upper triangular and U unitary.

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Since $A = A^*$, $A = \overline{(UTU^{-1})}^T$

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So $T = \overline{T}^T$. And so T must be diagonal with real entries. ■

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“A real symmetric matrix is orthogonally diagonalizable.”

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Singular Value Decomposition
 $A= Q^{-1} D P$, with Q & P orthogonal

matrix equivalence

$$\mathbb{R}^3 \rightarrow \mathbb{R}^2$$

$$\begin{pmatrix} 1 & 0 & 5 \\ 2 & 4 & 6 \end{pmatrix}$$

matrix equivalence

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$$\begin{pmatrix} 1 & 0 & 5 \\ 2 & 4 & 6 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 0 \\ -2 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 5 \\ 0 & 4 & -4 \end{pmatrix}$$

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$$\begin{pmatrix} 1 & 0 \\ 0 & 1/4 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ -2 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 5 \\ 0 & 1 & -1 \end{pmatrix}$$

matrix equivalence

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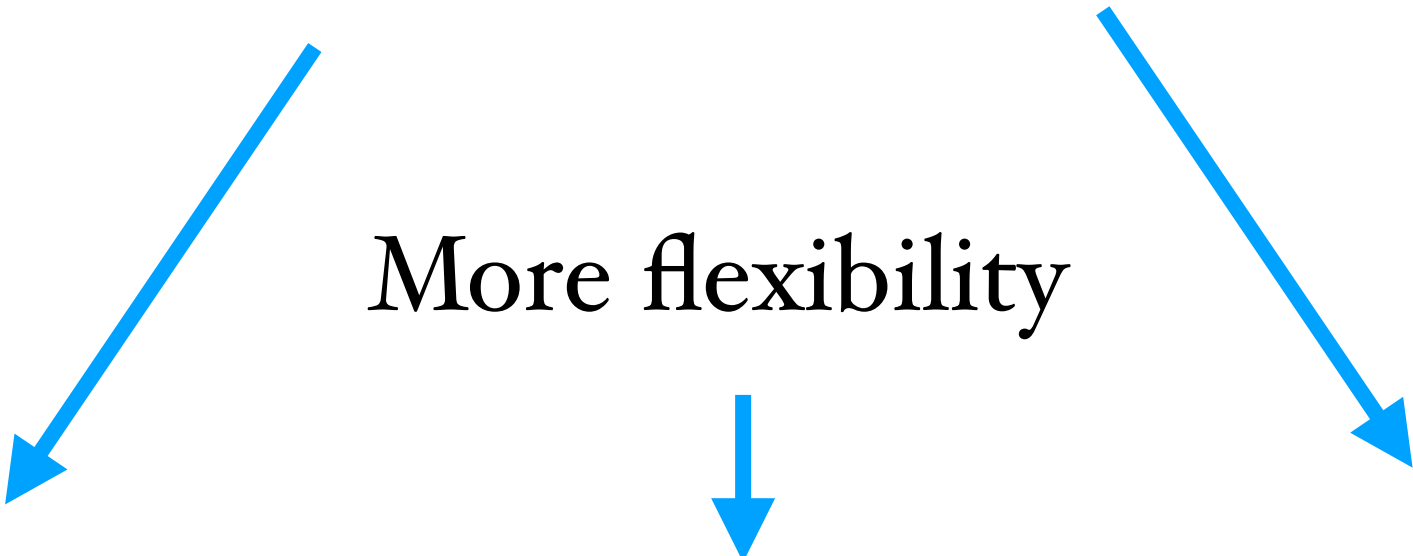
$$\begin{pmatrix} 1 & 0 \\ 0 & 1/4 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ -2 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 5 \\ 0 & 1 & -1 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 0 \\ 0 & 1/4 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ -2 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix} \begin{pmatrix} 1 & 0 & -5 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 0 & 5 \\ 2 & 4 & 6 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ -2 & 1/4 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix} \begin{pmatrix} 1 & 0 & -5 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix}$$

Less flexibility

More flexibility



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matrix equivalence:
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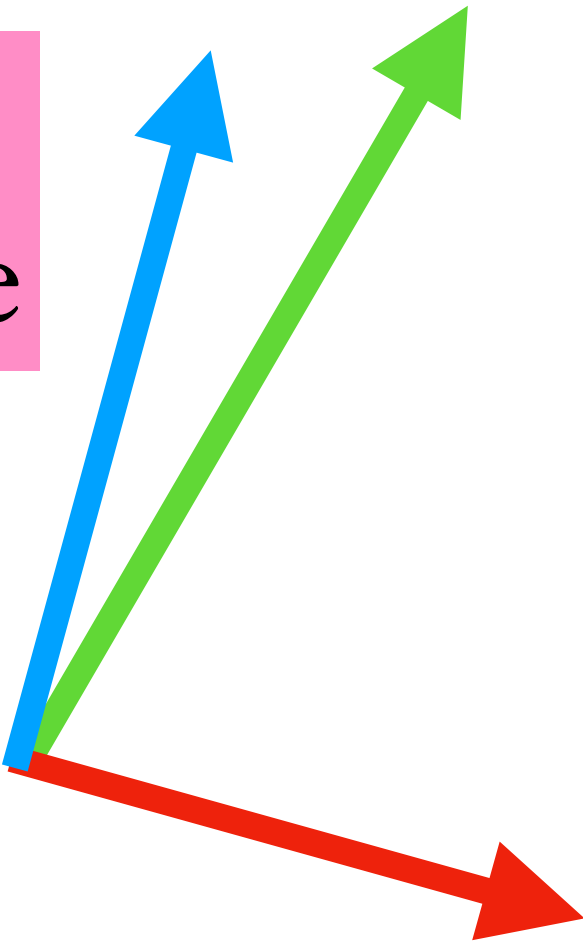
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 $A = PDP^{-1}$ with real entries for D and
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Singular Value Decomposition
 $A= Q^{-1}P$, with Q & P unitary (orthogonal)

matrix
equivalence



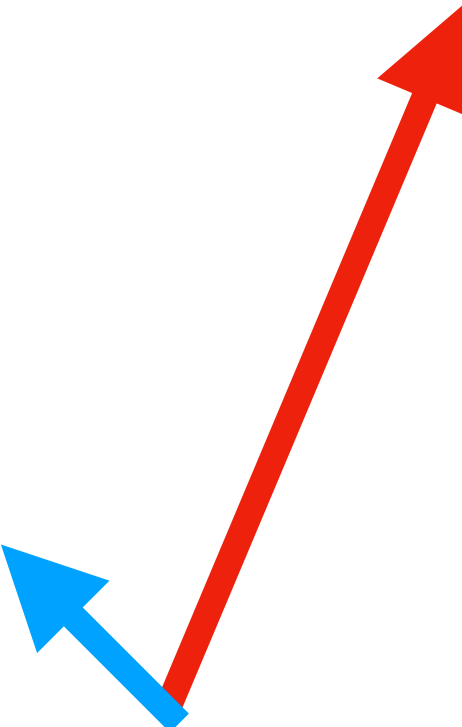
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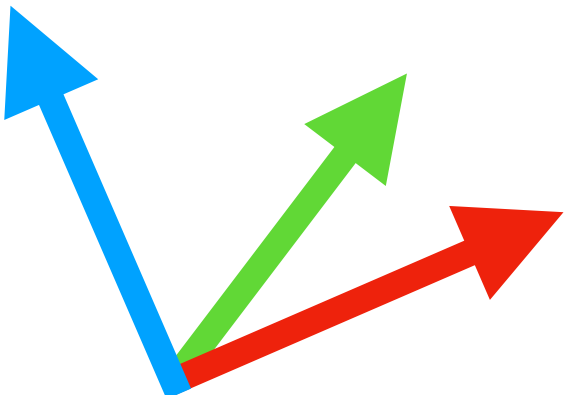
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Q



Singular Value
Decomposition

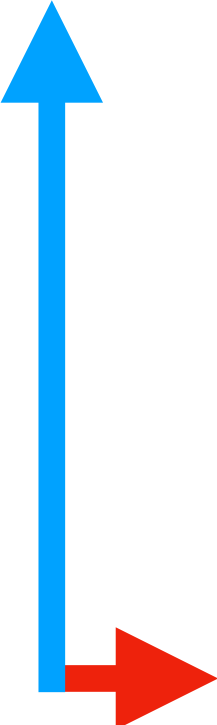


P

(unitary/
orthogonal)

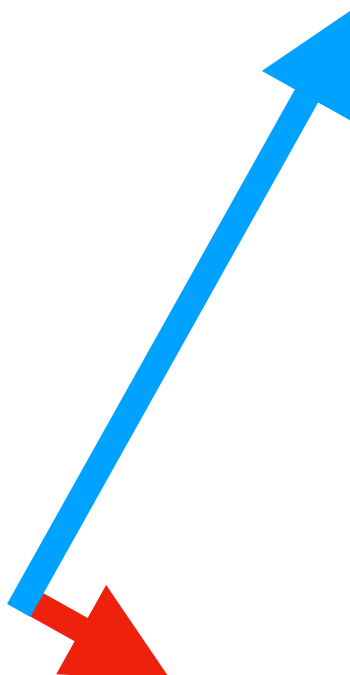


“ D ”



Q

(unitary/
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